Ionic-Conductance-Enhanced Nanofluidic Memristor Enabling Low-Voltage Neuromorphic Computing

[1]Sungyeol Kwak†, [2]Dong-yup Lee†, [3]Hyomin Lee, [1]Hayoung Yun, [2]Jeong-Yun Sun and [1,4,5,6]Sung Jae Kim*
[1]Department of Electrical and Computer Engineering, Seoul National University (SNU), [2]Department of Material Science and Engineering, SNU, [3]Department of Chemical Engineering and Green Energy, [4]Seoul National University Energy Initiative (SNUEI), [5]SOFT Foundry Institute, SNU, [6]Inter-university Semiconductor Research Center, SNU
E-mail: gates@snu.ac.kr

이온 전도도 향상 나노유체 멤리스터 기반 저전압 뉴로모픽 컴퓨팅

[1] 곽성열, [2] 이동엽, [3] 이효민, [1] 윤하영, [2] 선정윤 [1,4,5,6] 김성재*,
[1] 전기정보공학부, 서울대학교 [2] 재료공학부, 서울대, [3] 화공에너지그린학과, 제주대,
[4] 에너지이니셔티브연구단, 서울대, [5] 소프트파운드로연구소, 서울대, [6] 반도체공동연구소, 서울대

Abstract

Fluid memristors have emerged as promising neuromorphic devices by emulating ion-mediated signal processing in biological synapses. However, most reported memristors still operate at relatively high voltages and employ simplified device architectures, limiting their applicability to low-voltage neuromorphic systems and direct interaction with bio-potential-level signals. Here, we present a nanofluidic memristor based on a three-dimensional asymmetric architecture integrated with nanoporous hydrogel PN junctions. By exploiting coupled electric, diffusive, and electrohydrodynamic ion transport, the device exhibits nonlinear memory behavior and synaptic plasticity under genuine bio-potential-level stimuli. The hydrogel-based architecture further enables intrinsic biocompatibility and ultra-low-voltage operation, establishing nanofluidic memristors as a viable platform for low-power neuromorphic computing.

Keywords: *Neuromorphic*(뉴로모픽), *Nanofluidic*(미세유체), *Memristor*(멤리스터), *Hydrogel*(하이드로젤)

1. Introduction

A memristor is a nonlinear resistive device whose conductance depends on the history of applied electrical stimuli. Originally proposed as the fourth fundamental circuit element following resistors, capacitors, and inductors (Fig. 1-(a)), it intrinsically couples memory and computation through history-dependent conductance modulation, enabling compact and energy-efficient neuromorphic hardware [1]. In biological neural networks, synaptic plasticity refers to the activity-dependent modulation of synaptic strength and forms the basis of learning and memory. Neuromorphic devices emulate this behavior via gradual and history-dependent conductance changes, allowing potentiation, depression, and short- and long-term plasticity (STP and LTP) (Fig. 1-(b)). However, most solid-state memristors operate at relatively high voltages and lack compatibility with bio-potential-level signals [2]. Recently, liquid and nanofluidic memristors have emerged as promising alternatives by exploiting ion-mediated transport mechanisms analogous to biological synapses [3]. Here, we introduce a nanofluidic memristor based on a three-dimensional asymmetric architecture with nanoporous hydrogel PN junctions, enabling ultra-low-voltage synaptic plasticity.

2.1 Fabrication of a hydrogel-based fluidic memristor with improved ionic conductance

To reduce energy consumption and enable operation with biologically derived signals, memristors must be driven at low voltages. In such low-voltage regimes, conductance modulation critically depends on sufficient ionic motion within the active medium, which requires a relatively high ionic conductance. When ionic transport is limited, stronger electric fields are needed to induce ion migration, leading to increased operating voltage and power consumption. a highly charged ionic hydrogel is employed to promote ionic motion under weak electric fields. The high fixed-charge density of the hydrogel increases the concentration of mobile counter-ions, thereby supporting ion transport and conductance modulation even at reduced voltages (Fig. 2-(a)). In addition, a three-dimensional fluidic configuration is introduced to spatially confine and concentrate the electric field along the ionic transport pathways (Fig. 2-(b)). By directing the electric field toward regions where ion migration predominantly occurs, effective ionic motion can be achieved without increasing the applied voltage.

2.2. Neuromorphic behavior of the proposed memristor device

The neuromorphic properties of the proposed memristor were investigated through electrical characterization. The device exhibits typical memristive fingerprints, including pinched hysteresis loops crossing the origin in the current–voltage characteristics (Fig. 3-(a)). In addition, the hysteresis loop area decreases as the excitation frequency increases, indicating that the conductance modulation is governed by ion migration dynamics with limited response at shorter timescales. Synaptic plasticity was evaluated using voltage pulse stimulation. A series of positive voltage pulses (+1 V, 1 s) induces a gradual increase in conductance, corresponding to synaptic potentiation, whereas repeated negative pulses (−1 V, 1 s) lead to a progressive decrease in conductance, representing synaptic depression (Fig. 3-(b)). The gradual and bidirectional conductance modulation demonstrates that the fluid memristor can emulate essential synaptic functions. Such analog and history-dependent conductance evolution originates from the cumulative redistribution of ions within the nanofluidic junction, providing a physical basis for synaptic weight update. These results

confirm that the proposed device reliably captures key features of biological synapses required for neuromorphic signal processing.

2.3 Low voltage operation and reservoir computing

The proposed device exhibits stable neuromorphic operation under genuine bio-potential-level stimuli. Short-term synaptic dynamics were successfully implemented with input amplitudes as low as 5 mV, indicating that nonlinear conductance modulation and temporal memory are preserved even under extremely weak electrical excitation (Fig. 4-(a)). Such characteristics are particularly well suited for reservoir computing, where dynamic nonlinearity and fading memory dominate computational performance. To evaluate its computational capability, a preliminary reservoir computing demonstration was performed using the MNIST handwritten digit dataset (Fig. 4-(b)). Input patterns were encoded using a write voltage of 5 mV, and the dynamic current responses of the device were used as reservoir states. Despite the ultra-low operating voltage, the system achieved a classification accuracy of 98%, comparable to previously reported memristor-based reservoir computing systems operating at significantly higher voltages. These results highlight the potential of the proposed device as an energy-efficient platform for low-voltage neuromorphic computing and bio-interfacing applications.

3. Conclusions

The results of this study highlight the feasibility of fluid memristors operating in a regime that closely matches biological signal amplitudes and timescales. The combination of a three-dimensional asymmetric architecture and nanoporous hydrogel PN junctions enables ionic transport dynamics that are difficult to achieve in conventional solid-state or previously reported fluid memristors, particularly under few mV excitations. Beyond demonstrating individual device functionality, this work suggests a new design strategy for neuromorphic hardware that prioritizes ultra-low energy consumption, and bio signal-level compatibility with physiological systems. The ability to support learning-related functions at few m *Nature* (2008) he applicability of fluid memristors from proof-of-concept devices toward practical neuromorphic and bio-interfaced computing platforms. These findings position hydrogel-based fluid memristors as a promising building block for future low-power, ion-based neuromorphic systems beyond the limitations of conventional electronic memristors.

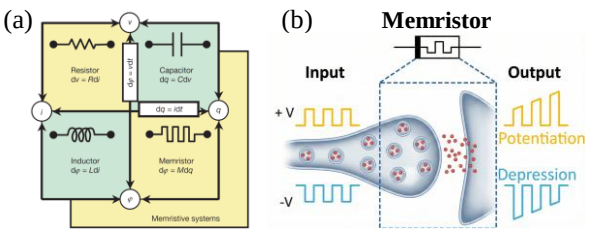

Fig. 1. Concept of memristor and synaptic plasticity. (a) Memristor as the fourth fundamental circuit element, exhibiting history-dependent conductance that couples memory and computation. (b) Emulation of biological synaptic plasticity in neuromorphic devices.

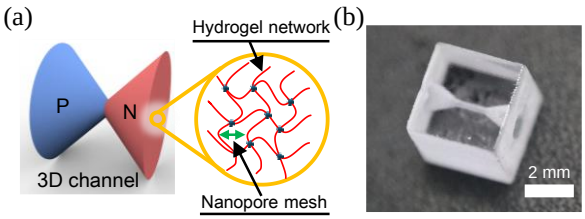

Fig. 2. Strategies for enhancing ionic conductance in the nanofluidic memristor Schematic of a high–fixed-charge nanoporous hydrogel enabling stable low-voltage conductance modulation. (b) Three-dimensional fluidic architecture designed to spatially confine and concentrate the electric field along the ionic transport pathways, enhancing low-voltage operation.

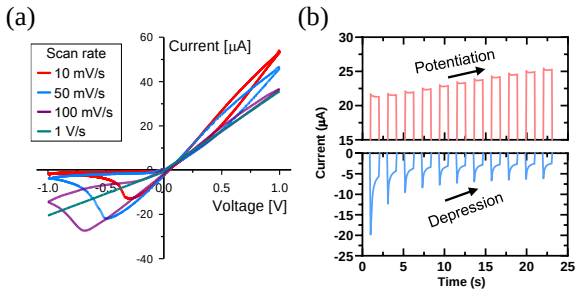

Fig. 3. Neuromorphic characteristics of the nanofluidic memristor.
(a) Current–voltage characteristics showing pinched hysteresis loops crossing the origin, with a decreasing loop area as the excitation frequency increases. (b) Synaptic plasticity under voltage pulse stimulation, where repeated positive voltage pulses (+1 V, 1 s) induce gradual conductance potentiation, while repeated negative pulses (−1 V, 1 s) result in conductance depression.

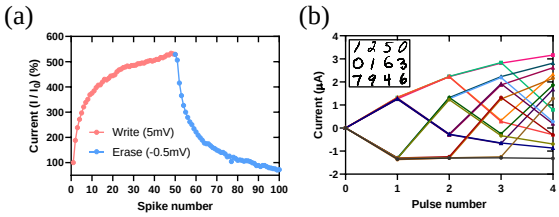

Fig. 4. Low-voltage neuromorphic operation and reservoir computing demonstration. (a) Short-term plasticity of the proposed memristor under bio-potential-level stimulation, showing nonlinear conductance modulation and temporal memory at input amplitudes as low as 5 mV. (b) Preliminary reservoir computing demonstration using the MNIST handwritten digit dataset, validating the suitability of the device for dynamic and nonlinear temporal processing.

Acknowledgments

This work is supported by BK21 PLUS program and Carbon Neutral Campus Initiative, 2025 Hyper-Convergence Research Support Program (0681-20250036) at Seoul National University.

References

1. Strukov, D. B., et al. “The Missing Memristor Found.” Nature, vol. 453, no. 7191, May 2008, pp. 80–83.
2. B. Govoreanu et al., "10×10nm² Hf/HfO_x Crossbar Resistive RAM with Excellent Performance, Reliability and Low-Energy Operation," 2011 International Electron Devices Meeting (IEDM), Washington, DC, USA, 2011, pp. 31.6.1-31.6.4.
3. Robin, P., et al. “Long-Term Memory and Synapse-like Dynamics in Two-Dimensional Nanofluidic Channels.” Science, vol. 379, no. 6628, Jan. 2023, pp. 161–67.